Reactive Routing Using AODV in Wireless Multi-Hop Ad Hoc Networks

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***Abstract*—Wireless multi-hop ad hoc networks are collections of mobile nodes that communicate without relying on fixed infrastructure, either directly or through intermediate nodes that forward packets over multiple hops. Since links may change frequently in such networks, routing protocols must adapt to topology changes while keeping routing overhead low. This paper examines the Ad hoc On-Demand Distance Vector (AODV) protocol as a reactive routing protocol for wireless multi-hop ad hoc networks. It describes how AODV discovers routes using Route Request (RREQ) and Route Reply (RREP) messages, how Expanding Ring Search can help reduce RREQ flooding, and how Route Error (RERR) messages support route maintenance after link failures. The paper also explains important AODV control message fields, routing table entries, Hello messages, and the role of sequence numbers in avoiding stale route information and supporting loop-free routing. Finally, the paper compares AODV with proactive routing and Dynamic Source Routing (DSR), discusses the trade-off between low routing overhead and possible delay during route discovery and briefly addresses applications and later extensions of AODV.**

I. Introduction

Wireless multi-hop ad hoc networks enable communication between mobile nodes without relying on fixed infrastructure such as base stations or access points. They are called multi-hop networks because data may be forwarded through intermediate nodes when the source and destination are not within direct transmission range. Nodes may therefore communicate directly when they are within transmission range, or indirectly through intermediate nodes that forward packets [1], [2]. This makes such networks useful in situations where communication must be established quickly or where fixed infrastructure is unavailable [1]. However, the absence of a fixed infrastructure also makes routing more difficult, because the network topology can change as nodes move or links become unavailable [2].

A central challenge in such networks is the balance between route availability and routing overhead. Proactive routing protocols keep routes available by maintaining routing information continuously, but this can consume bandwidth and energy even when no data is being transmitted. on the other hand reactive routing protocols follow a different approach, they discover routes only when communication is needed [3]. This can reduce unnecessary control traffic, but it also may introduce delay before data transmission can begin [2].

Ad hoc On-Demand Distance Vector (AODV) is a reactive routing protocol for mobile ad hoc networks [4], [5]. It discovers routes on demand using Route Request (RREQ) and Route Reply (RREP) messages, also it maintains active routes using mechanisms such as Route Error (RERR) messages, and uses destination sequence numbers to select fresh routes and support loop-free routing [6].

This paper explains the operation of AODV in wireless multi-hop ad hoc networks. It first introduces the background of ad hoc routing and then describes the main AODV mechanisms used for route discovery, route maintenance, and loop-free routing. Finally, it compares AODV with proactive routing and Dynamic Source Routing (DSR), discusses applications and extensions of AODV and summarizes its main advantages and limitations.

II. Background

Wireless ad hoc networks are a concrete example of wireless multi-hop networks. In such networks, each node is not only a sender or receiver of data, but may also forward packets for other nodes. Multi-hop forwarding is needed when direct single-hop communication is not possible, for example because the destination is outside the source node's transmission range. Routing protocols must therefore find paths over intermediate nodes and adapt these paths when mobility or link changes make a previous route unusable [2].

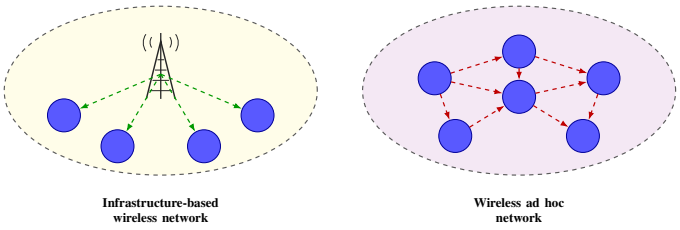


Figure 1. Conceptual comparison between an infrastructure-based wireless network and a wireless ad hoc network.

As shown in Fig. 1, the basic difference between an infrastructure-based wireless network and a wireless ad hoc network can be illustrated by comparing their communication structures. In an infrastructure-based network, communication

relies on fixed network components. In contrast, in a wireless ad hoc network, there is no fixed infrastructure, and autonomous nodes communicate directly with one another. If direct communication between two nodes is not possible, data may be forwarded through intermediate nodes over multiple hops. This decentralized and multi-hop communication structure is one of the characteristics of ad hoc networks and forms the basis for routing protocols such as AODV [2].

This dynamic topology makes routing more difficult than in fixed networks. A route that is valid at one point in time may become unusable (expired) shortly afterwards due to mobility, interference, or link failure [2]. As a result, routing protocols must react to changes while avoiding excessive control traffic. Routing overhead is important in this context because control messages consume bandwidth and energy that would otherwise be available for data transmission [2].

Routing protocols for ad hoc networks are commonly divided into proactive and reactive approaches. Proactive protocols maintain routing information continuously, so routes are usually available when data transmission begins [3]. However, this requires periodic updates and can create overhead even when routes are not used. Reactive protocols, by contrast, discover routes only when they are needed. This reduces unnecessary route maintenance, but it can introduce an initial delay because a route must first be found before packets can be sent [3].

AODV follows the reactive approach. It does not maintain routes to destinations that are not in active communication. Instead, it discovers a route when a source node needs to communicate with a destination for which no valid route is available [6]. This makes AODV suitable for dynamic ad hoc networks, but its performance depends on how often routes break and how much control traffic is needed to repair or rediscover them [4], [6].

III. Operation of AODV

The operation of AODV can be divided into three connected phases: route discovery, route use, and route maintenance. Route discovery is responsible for finding a path only when communication is required. Route use relies on routing table entries that store next-hop information for forwarding packets. Route maintenance reacts to link failures by invalidating broken routes and if necessary supporting local repair or a new route discovery process. In addition, AODV relies on destination sequence numbers to keep routing information fresh and to support loop-free routing. These mechanisms are explained in the following subsections [6]

A. Route Discovery and Expanding Ring Search in AODV

Route Discovery.

The route discovery phase in AODV begins when a source node needs to transfer data to a particular destination, but the route to that destination is not present in its routing table. Unlike proactive routing approaches, AODV creates routes only when needed rather than keeping complete routing information for all destinations at all times. As first step in the route discovery process, the source node broadcasts a Route Request (RREQ) message containing important fields such as the RREQ ID, destination IP address, destination sequence number, originator IP address, and originator sequence number [6].

After the RREQ is constructed, the message is broadcasted to neighboring nodes so that the route discovery process can propagate through the network. When an intermediate node receives the RREQ, it checks whether the request has already been processed, duplicate RREQs are discarded. For a non-duplicate RREQ, the node then creates or updates a reverse route to the originator of the RREQ. This reverse route allows a Route Reply (RREP) to later be sent back toward the source. If the node is not the destination and does not have a valid and sufficiently fresh route to the destination, it increments the hop count and rebroadcasts the RREQ to its neighbors. In this way, the RREQ continues to propagate through the network until it reaches either the destination node or an intermediate node that can satisfy the request [6].

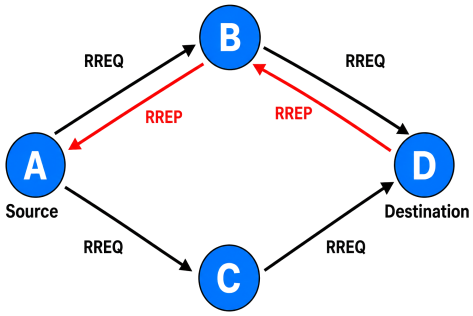


Figure 2. Illustration of route discovery in AODV. The RREQ is broadcast from the source, while the RREP returns along one selected reverse path.

As shown in Fig. 2, the route discovery process can be illustrated with node A as the source, nodes B and C as intermediate nodes, and node D as the destination. The figure shows how the RREQ spreads from A through neighboring nodes until D, or an intermediate node with a valid and sufficiently fresh route, can answer the request. The RREP is then sent back along one selected reverse path, for example from D through B back to A. During this return process, the nodes on the path create forward route entries toward the destination. Once the RREP reaches node A, the route is established and data transmission can begin [6].

Route Request (RREQ).

The RREQ ID allows receiving nodes to check whether they have already received and processed the same RREQ from the same source. However, considering the RREQ ID alone would not be sufficient, since different nodes may use the same identifier value. Therefore, it is necessary to consider both the RREQ ID and the IP address of the node originating the request. This combination allows each route request to be identified uniquely.

The destination IP address identifies the final destination, so it simply tells the network which node the route is being searched for. The destination sequence number indicates the freshness of the route information known by the originator.

The originator IP address and the originator sequence number identify the source of the request. This allows intermediate nodes to create a reverse route back to the source so that a later Route Reply (RREP) can be returned in the correct direction. The originator sequence number also helps indicate the freshness of the route information related to the source node.

The hop count field shows how many hops the RREQ message has travelled from the originator to the node currently handling the request. As a result, it gives the distance between the source and the current node in terms of hops [6].

The RREQ message may also contain flags that control how the route discovery process should behave. One important example is the destination-only flag, also called the D flag. If this flag is set, only the actual destination node is allowed to generate a Route Reply. This means that intermediate nodes are not allowed to answer the request, even if they already have a route to the destination.

Another example is the gratuitous RREP flag, also called the G flag. This flag is used when an intermediate node replies to a route request and the destination should also be informed about the route back to the source. In this case, a gratuitous RREP can be sent toward the destination. This can be useful when the destination also needs route information back to the originator [6].

Route Reply (RREP).

When the RREQ reaches the destination node or an intermediate node with a valid and sufficiently fresh route to the requested destination, an RREP message is generated. An intermediate node should only answer if the destination sequence number in its routing table is greater than or equal to the destination sequence number requested in the RREQ. Otherwise, the node does not send an RREP and the RREQ continues to propagate through the network. The destination sequence number in the RREP shows the freshness of the route information.

The originator IP address identifies the source node that the RREP is sent back to along the reverse route. The lifetime field indicates how long the route should be considered valid. The hop count field shows the number of hops on the route to the destination.

The RREP message format also contains an acknowledgment flag, called the A flag. This flag can be used when confirmation is needed that the RREP was successfully received, for example in situations where a link may be unreliable or unidirectional [6].

Expanding Ring Search.

During route discovery, a complete network-wide broadcast of Route Request (RREQ) messages may create unnecessary overhead, especially when the destination can be found within a small Time To Live(TTL) range or an early search ring. In such a case, many nodes may participate in forwarding control messages even though a shorter search would have been sufficient. This consumes bandwidth and energy that could otherwise be used for actual data transmission [7]. For this reason, AODV applies a mechanism known as Expanding Ring Search in order to limit the search area during the early stages of route discovery [6].

This mechanism uses the TTL field in the IP header to control how far an RREQ message may propagate through the network. Initially, the source node begins route discovery with a relatively small TTL value so that only nearby nodes participate in the search. If no Route Reply (RREP) is received, the source node repeats the discovery procedure with a larger TTL value [6]. Thus, the search area is gradually expanded step by step instead of allowing the request to spread immediately throughout the entire network [6].

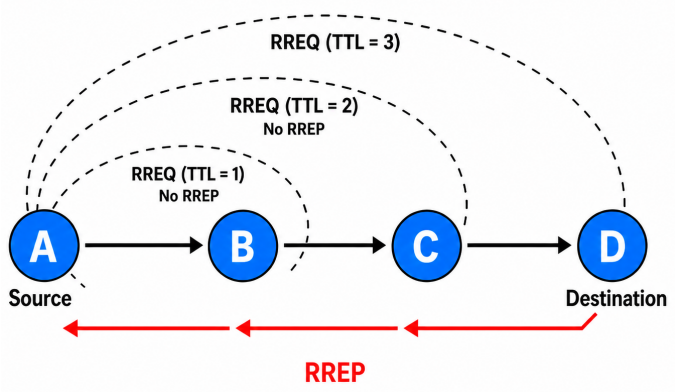


Figure 3. Illustration of Expanding Ring Search in AODV using progressively increasing TTL values.

As shown in Fig. 3, Expanding Ring Search allows the source node to begin route discovery with a small TTL value and to increase this value only when no RREP is received. In the illustrated example, the first two searches do not reach the destination, while the search with TTL = 3 succeeds and the destination returns an RREP to the source.

The main advantage of this mechanism is that it can reduce unnecessary routing traffic and support more efficient energy use in wireless nodes [7]. It can also help with limiting large scale RREQ flooding in AODV [6]. Such flooding is related to the broadcast storm problem in MANETs, where simple flooding can cause redundant rebroadcasts, contention, and collisions [8]. However, if the destination is actually far away, the source node may need several attempts with increasing TTL values before a route is discovered [6]. As a result, Expanding Ring Search can reduce overhead, but if the destination is not found in the early search rings, it may introduce additional delay during route discovery. This reflects the same balance that is central to reactive routing in general: lower control overhead at the cost of possible route discovery delay [3], [6].

B. Route Maintenance and Error Handling in AODV

General Route Maintenance.

After a route has been established and is being used for data transmission, its validity must be maintained because ad hoc networks are highly dynamic environments. A previously valid route may become unusable due to node movement or the loss of a link between neighboring nodes. For this reason, route maintenance is an essential procedure in AODV, in order

to monitor active routes, detect link failures, notify affected nodes, and support continued communication either through local repair or by initiating a new route discovery process [6].

To maintain connectivity to the next hop in an active route, nodes may use different mechanisms, such as link-layer notifications or Hello messages exchanged between neighbors. If a node can no longer confirm connectivity to the next hop, it assumes that the link has been lost. As a result, the route can no longer be relied upon, and route maintenance procedures must be initiated. In this way, AODV checks whether active routes remain usable instead of assuming that previously discovered routes will stay valid indefinitely [6].

Hello Messages

Unlike RREQ, RREP, and RERR, the Hello message is not a separate AODV message format. Instead, it is sent as a RREP with TTL = 1. Thsu, a Hello message remains within one hop and reaches only the immediate neighboring nodes.

They are used in AODV to determine whether neighboring nodes and next hops are still reachable. If a node receives Hello messages or other packets from a neighbor, it can use this information to maintain local connectivity. On the other hand, if no packets, including Hello messages, are received from a neighbor for a certain period of time, the node may assume that the link to this neighbor has been broken.

The interval at which Hello messages may be sent is defined by the HELLO_INTERVAL value. However, Hello messages are not mandatory in all situations, they are mainly useful when a node needs to check the reachability of neighbors on active routes. If other mechanisms, such as link-layer feedback, are available to detect link failures, Hello messages may not be necessary.

The disadvantage of Hello messages is that they are sent periodically and therefore create additional control traffic, even though they do not carry actual user data. For this reason, AODV allows link failures to be detected either through Hello messages or through link-layer notifications [6].

Route Error (RERR)

A Route Error (RERR) message is generated when a node detects that an active route is no longer valid, which may happen for example, when a node can no longer reach the next hop on the route, when a link break is detected while transmitting data, or when Hello messages from a neighboring node are no longer received. In such a case, the node informs the affected nodes that the previous route can no longer be used.

A RERR message may contain more than one unreachable destination. This is useful because a single link break can affect several routes at the same time. For example, if one next hop provides access to multiple destinations, the failure of that next hop makes all of those destinations unreachable over the previous path. Encoding this information in a single RERR message avoids the need to send separate error information for each destination.

In the RERR message, each unreachable destination is associated with a destination sequence number. This value allows other nodes to determine whether the information about the broken route is newer than the information already stored in their routing tables. Without it, a node might incorrectly continue to use an invalid route. The sequence number therefore helps nodes invalidate stale route entries and avoid the use of broken routes.

After the unreachable destinations have been identified, the RERR message is not sent to all nodes in the network. Instead, AODV uses precursor information to determine which neighboring nodes should be informed. The precursor list contains the neighboring nodes that use this node as their next hop toward the destination. In this way, the error message is forwarded only to nodes that are affected by the link failure.

The RERR message may also include the No-Delete flag, also called the N flag. This flag is related to local repair. It indicates that upstream nodes should not immediately delete the route entry, because a local repair attempt may still restore connectivity. If the repair succeeds, the route can be updated. If the repair fails, the route is treated as invalid and a new route discovery may be necessary [6].

Route Maintenance Example

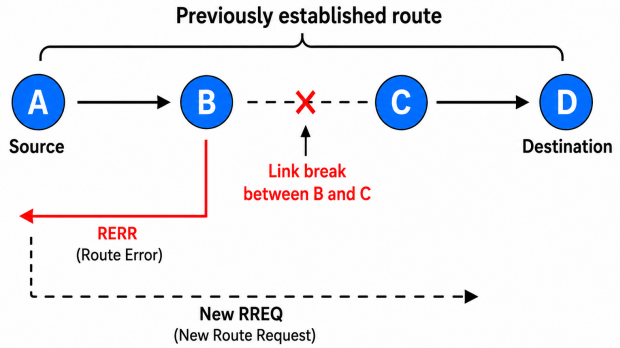


Figure 4. Illustration of AODV route maintenance after a link failure.

As shown in Fig. 4, route maintenance in AODV can be illustrated with a previously established route from node A to node D through the intermediate nodes B and C. If the link between B and C breaks, the route becomes invalid and packets can no longer be forwarded along the same path. In this situation, node B, as the upstream node close to the broken link, may detect the link failure, for example through missing Hello messages or link-layer feedback.

After detecting the broken link, node B may attempt local repair if the destination is still within a suitable repair range [6]. If local repair is not attempted or fails, node B generates a Route Error (RERR) message to notify affected nodes that the previous route to the destination is no longer valid. The affected routes are then marked as invalid, and the corresponding unreachable destinations are identified. AODV uses precursor information to determine which neighboring nodes should be informed about the broken route. If communication with node D is still required, node A may start a new route discovery process by sending a new RREQ. Thus, route maintenance enables AODV to react to link failures, remove invalid routes, and restore communication through local repair or renewed route discovery [6].

C. Routing Tables, Sequence Numbers and Loop Freedom

Routing Table Entries

AODV stores routing information in routing tables at the nodes because it uses hop-by-hop routing instead of placing the complete path inside each data packet, as in source routing protocols such as DSR. Thus, each node only needs to know the next hop toward a destination, rather than the full route. This can reduce packet-header overhead compared with source routing, especially when the path contains many hops. Therefore, each entry in the routing table represents the current knowledge of a node about how to reach a particular destination [6], [9].

As in AODV control messages, the destination IP address and destination sequence number are also stored in routing table entries to identify the destination and indicate the freshness of the stored route information.

The valid destination sequence number flag indicates whether the stored destination sequence number is known and valid. This is important because in some cases, a node may have an entry for a destination even when the sequence number cannot be reliably used for route comparison. In contrast, the route state indicates whether the route itself can currently be used for packet forwarding. If a link break is detected or an RERR message is received, the route may be marked as invalid and should no longer be used for data transmission.

The hop count and next hop fields describe how the destination can be reached from the current node. The hop count gives the path length, while the next hop identifies the neighboring node to which packets should be forwarded.

The precursor list stores neighboring nodes that depend on this routing entry. This information is later used when a route becomes invalid and affected nodes need to be informed. Finally, the lifetime or expiration time specifies how long the entry remains valid. When this lifetime expires, the route should no longer be used as an active route. If the node still needs to communicate with the destination, a new route discovery process may be required.

Thus, AODV messages and routing table entries are closely related. RREQ and RREP messages are used to create or update routing entries during route discovery, while RERR messages are used to invalidate routing entries when routes break. In this way, the messages represent the operational mechanism of route creation and maintenance, while the routing table stores the results of this process [6].

Sequence Numbers and Loop Freedom

In a wireless ad hoc network, AODV uses sequence numbers to check the freshness of routing information for specific destinations. The importance of sequence numbers comes from the nature of wireless ad hoc networks, where route information may become outdated because of node movement and changes in network topology. Thus, AODV uses destination sequence numbers to distinguish between older and newer routes. Sequence numbers are stored in routing table entries and are also included in AODV routing messages used during the route discovery process. In general, a higher destination sequence number indicates fresher routing information. When a node discovers two alternative routes to the same destination, the preferred route is the one with the higher destination sequence number, but if the two routes have an equal destination sequence number, the one with the lower hop count is chosen [6].

Stale routes are a problem in ad hoc networks due to frequent changes in routing information. For instance, a route that worked before may become invalid after a change in network topology [2]. If stale routing information is used, packets may be sent toward unreachable nodes or invalid next hops, which can lead to packet loss or require a new route discovery process. Also, AODV needs a mechanism that helps select fresh routes whenever they are available and avoids using stale routes whenever possible [6].

furthermore, sequence numbers enable AODV to support loop-free routing. A routing loop is a situation in which packets continuously circulate among nodes without reaching the destination. One possible cause of routing loops is that nodes make forwarding decisions based on outdated or incorrect route information. And by preferring routes with higher destination sequence numbers, AODV lowers the chance of selecting stale routes that could otherwise contribute to routing loops [6].

IV. Discussion and Comparison of AODV

A. Brief Comparison with Other Routing Approaches

The difference between AODV and proactive routing lies mainly in the way routes are maintained. While proactive routing continuously maintains up-to-date routes to all other nodes regardless of whether data exchange is taking place, AODV searches for a route only when there is a need to transmit data [3], [6].

Another difference between AODV and another reactive routing protocol such as Dynamic Source Routing (DSR) lies in how route information is stored and used for packet forwarding. AODV uses routing tables stored at nodes and destination sequence numbers, while DSR uses source routing, where the route is carried in the packet header. This means that AODV avoids carrying complete path information in every data packet, which can be useful when smaller packet headers are preferred [6], [9].

Table I summarizes the main differences between proactive routing, AODV, and DSR in terms of route creation, control overhead, and initial delay [3], [6], [9].

B. Performance Criteria Used to Evaluate AODV

The evaluation of AODV should not be limited to whether the protocol functions or not, but it should consider several performance criteria that show how effectively the protocol operates under different network conditions. In general, evaluating a routing protocol means examining not only whether it can discover valid routes, but also how reliably it delivers packets, how much delay it introduces, how much routing overhead it generates, and how well it performs in both stable and highly dynamic network conditions [2], [10].

One important criterion is the packet delivery fraction, which indicates how many packets successfully reach their final destination compared with the total number of packets

Table I. COMPARISON OF PROACTIVE ROUTING, AODV, AND DSR IN MANETS

Protocol	Route creation	Control overhead	Initial delay
Proactive	Routes maintained continuously.	High due to periodic updates.	Low (routes are usually available).
AODV	Routes created on demand (next-hop entries and sequence numbers).	Low to moderate(increases with frequent rediscovery).	Moderate(route discovery is needed first).
DSR	Routes created on demand (source routing and route caches).	Low periodic overhead, but larger packet headers.	Moderate (cached routes may reduce delay).

generated by the source [10]. This metric is especially important in ad hoc networks because frequent route changes, heavy traffic, and repeated link breaks can make reliable delivery difficult. In AODV, a high packet delivery fraction suggests that the protocol can discover appropriate routes, maintain connectivity, and deliver packets effectively. In contrast, a low packet delivery fraction may indicate problems caused by frequent route failures, delays in route rediscovery, excessive routing overhead, or packet loss caused by topology changes [2], [10]. Another important criterion is end-to-end delay, which refers to the total time required for a packet to travel from the sender to the destination. In AODV, this delay may increase because routes are not maintained in advance. Instead, the protocol first has to initiate route discovery by sending an RREQ and waiting for an RREP before actual data transmission can begin. Delay may also rise when an already established route breaks and a new one must be discovered [6], [10].

Routing overhead, often measured as normalized routing load, is another major criterion used to evaluate AODV. It refers to the amount of network resources consumed by the protocol for route discovery and route maintenance instead of actual data transmission [2], [10]. In AODV, control packets such as RREQ, RREP, and RERR messages, as well as Hello messages used for maintaining local connectivity, all contribute to routing overhead [6]. High routing overhead means that more bandwidth is spent on control activity, which can reduce the available resources for useful data transmission [2].

Throughput is also important, since it measures the amount of data successfully delivered over the network in a given period of time [2], [10]. High throughput indicates efficient data delivery, while frequent route breaks, repeated rediscovery procedures, and high control traffic may reduce throughput by increasing delay or causing packets to be dropped. Overall, these criteria provide a broader view of AODV performance. They show that AODV performance depends not only on whether routes can be discovered, but also on mobility, link failures, and the amount of control traffic generated during route discovery and maintenance [2], [6], [10].

C. Applications and Extensions of AODV

Although AODV was originally designed for mobile ad hoc networks [6], its basic principles are related to other wireless ad hoc network types, for example wireless sensor networks, wireless mesh networks, and vehicular ad hoc networks, which also involve changing topologies and multi-hop communication [11]. In dynamic multi-hop scenarios, the on-demand nature of AODV can be useful because routes are created only when communication is required, instead of maintaining unused routes continuously [6]. However, standard AODV is not automatically optimal for every wireless network type. In power-constrained or highly dynamic environments, route discovery delay, repeated rediscovery, and routing overhead may still become problematic, and for this reason later research has proposed several AODV extensions and modifications to improve aspects such as overhead, energy efficiency, routing performance, quality of service or security [4].

V. ADVANTAGES AND LIMITATIONS OF AODV

AODV routing is well suited for changing ad hoc wireless networks because it combines on-demand route creation with mechanisms for route freshness and loop freedom. Since AODV does not maintain routes to destinations that are not in active communication, it can reduce unnecessary control traffic and bandwidth usage. This is useful in dynamic networks where maintaining complete routing information at all times would consume additional resources. In addition, destination sequence numbers help nodes select fresher routes and avoid outdated routing information. As a result, AODV can react to topology changes while still reducing the risk of routing loops [5], [6].

One limitation of the AODV protocol is that route discovery may introduce additional delay because it is required before communication can begin. Specifically, a node cannot start transmitting data immediately if no valid route is available, since route creation requires sending an RREQ message and waiting for an RREP. This means that route discovery may take extra time, especially if the destination is not found quickly [3], [6].

In addition, if changes in the network happen too often, established routes may become invalid very quickly and nodes may need to discover new routes repeatedly, which results in a larger amount of control traffic [6]. Finally, frequent route discovery reduces efficiency because it consumes time, bandwidth, and other network resources while decreasing the available resources for actual data transmission [3], [6].

VI. CONCLUSION

Reactive routing is important in wireless ad hoc networks because routes are discovered on demand. The AODV routing protocol follows this reactive approach because it reduces unnecessary route maintenance while still supporting communication in dynamic network conditions. Routes are established using RREQ and RREP messages. Route maintenance is performed with mechanisms such maintaining local connectivity, RERR messages, local repair, and renewed route discovery after link failures.

The main advantage of AODV is that it combines on-demand route discovery with sequence numbers for selecting fresh routes and avoiding routing loops. However, it also has some limitations, route discovery may introduce delay before data transmission can begin. Furthermore, frequent topology changes can increase routing overhead because routes may need to be rediscovered repeatedly. The principles of AODV are also relevant for other dynamic wireless multi-hop environments, although later extensions are often needed to address aspects such as overhead, energy efficiency, quality of service, or security. Overall, AODV can work effectively in dynamic ad hoc networks, but its performance is strongly influenced by mobility, link stability, and the amount of routing overhead generated.

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